

F-65 Megawing

AI Behavior and Decision Architecture White Paper

Version 1.0 - Human-Reviewed Engineering Baseline - 25 August 2026

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Document Type: Technical / Product AI Engineering White Paper

Platform: MEGA65 - 45GS02, C-primary (LLVM-MOS) + measured 45GS02 assembly

Audience: Product owner, architecture, gameplay, engine implementers, C/ASM developers, mission authors, doctrine authors

Status: FINAL - HUMAN-REVIEWED v1.0 / APPROVED REWRITE INPUT

Date: 25 August 2026

Nature: Complete conceptual AI engineering baseline covering tactical decision-making, bounded perception, team doctrine, BVR/WVR combat, defensive behavior, IADS, mobility, navigation, procedural aircraft operations, ship movement, and ground-vehicle movement. Non-neural, deterministic, table/FSM driven, and designed for bounded execution on the MEGA65.

Approval effect: Human review is complete. Version 1.0 is the approved AI engineering baseline and rewrite input for the subsequent F-65 1.6 Architecture, Gameplay, Engine, Technical Alignment, and generated-interface work. It remains subordinate to the controlling specification hierarchy and does not by itself rewrite a parent document, freeze generated binary layouts, or promote later tuning/measurement values.

Authority and hierarchy posture

This paper is a **detailed subsystem engineering white paper** in the same posture as the Sensor / Radar white paper: a human-reviewed engineering baseline and approved rewrite input, **subordinate to the core three** of Architecture, Gameplay, and Engine, with Technical Alignment governing program control and propagation.

It does not silently become an additional controlling authority outside that hierarchy.

If the coordinated 1.6 corpus adopts this AI paper as the fifth detailed subsystem white paper alongside Physics, Graphics, Audio, and Radar/Sensors, the Main / Gameplay / Architecture / Engine rewrite shall list it explicitly and identify the required propagation.

White-paper concepts do not automatically become production features. Player-visible behavior introduced or refined here, notably Realistic / Novice AI modes, requires the applicable Gameplay / Main adoption described in Section 27.

Controlling parents

Role	Document
Architecture	F-65 Megawing Revision 1.5.1 - Architecture Invariants
Gameplay	F-65 Megawing Gameplay and Simulation Requirements Supplement
Engine	F-65 Engine Runtime and Toolchain Design Supplement Draft 0.2

Role	Document
Program control	F-65 Technical Alignment and Read-First Supplement v1.0
Sensor / tracks	F-65 Megawing SensorAndTrackEngine Engineering Model Phase-3 v1.0
Physics	MEGA65 Flight Simulation Physics 6DOF Atmosphere White Paper v3.3 - FROZEN
Graphics coexistence	F-65 Megawing Graphics White Paper v2.1
Audio coexistence	F-65 Megawing Audio, Sound Effects and Music Engineering White Paper v1.0

The SensorAndTrack Phase-3 v1.0 baseline closes TRACK-DOMAIN-001: the 24-semantic-track limit applies independently to each active tactical-network knowledge domain, currently MEGA / BLUE and RED.

This paper cannot relax architecture, timing, memory, ownership, deterministic-simulation, DMA, pool, resource-staging, sensor, physics, graphics, audio, or platform requirements established by its parents.

AI decisions execute only through tick-order stage 16. Commands and intents produced there become consumable no earlier than the following simulation tick.

AI never reads global truth.

Authoritative target AI uses fixed-width integer / fixed-point arithmetic only. No target floating-point arithmetic is permitted.

0. Approval and Relationship to Draft 0.9

Draft 0.9 was the final human-review candidate. Human review is now complete and the conceptual architecture is approved as Version 1.0.

Version 1.0 retains the full Draft 0.9 model:

- hierarchical doctrinal control;
- bounded perception and knowledge;
- HARD SAFETY and deterministic interrupt precedence;
- reflex versus doctrine evaluation;
- non-telepathic team coordination;
- BVR timeline and weapon-employment reasoning;
- defensive / missile-response logic;
- relational BFM;
- IADS and surface doctrine;
- authored-route mobility;
- aircraft parking-to-parking procedures;
- ship navigation;
- precompiled ground-vehicle navigation; and
- low-resource deterministic implementation.

The remaining items after conceptual freeze are implementation, schema, propagation, and measurement gates. They are recorded in Section 27 and **are not reasons to withhold Version 1.0 approval.**

1. Purpose and Scope

This paper establishes the practical contract for all AI-controlled entities in F-65 Megawing.

It covers:

- entity classification and doctrine composition;
- bounded perception and knowledge;
- hierarchical doctrinal control;
- deterministic safety and threat interrupts;
- reflex and deliberative decision cadence;
- mission and formation/team coordination;
- mobility, navigation, and procedural operations;
- aircraft cold/start-ready, taxi, takeoff, cruise, combat ingress, recovery, landing, rollout, and parking;
- ship navigation and tactical maneuver;
- ground-vehicle and mobile-SAM movement;
- BVR timeline logic and weapon-employment reasoning;
- defensive / missile-response behavior;
- WVR / BFM maneuver selection;
- IADS and surface doctrine;
- skill and proficiency variation without cheating;
- AI-to-engine intent boundaries; and
- deterministic low-resource implementation on the MEGA65.

Player aircraft remain under direct human control with RIO assistance.

All other aircraft, ships, SAM sites, applicable ground vehicles, and surface combatants are AI-driven according to their authored role.

Guided missiles after launch are not AI-controlled tactical entities. Their authoritative lifecycle belongs to WeaponAndDamageEngine.

2. Design Principles for Constrained Hardware

The following principles are mandatory.

1. **Hierarchical priority FSM** is the behavioral backbone.
2. **Interrupt precedence is deterministic and consequence-based.**
3. **Reflex and doctrine evaluation are separated** while remaining within stage 16.
4. **Utility scoring is local**, selecting only among doctrinally legal alternatives.
5. **Doctrine is composed** from bounded base tables and modifiers.
6. **Candidate sets are bounded.**
7. **Navigation uses authored/precompiled routes and simple guidance only.**
8. **No general-purpose runtime pathfinding.**
9. **Mission goals and ROE define the outer behavioral envelope.**
10. **No floating point in authoritative AI.**
11. **Behavioral variability may use only the entity-local AI RNG stream.**
12. **AI never bypasses owning simulation modules.**

No:

- neural models;
- GOAP;
- unrestricted behavior-tree expansion;
- recursive planning;
- A* during the sortie;
- general graph search during the sortie;
- continuous trajectory optimization;
- dynamic memory allocation; or
- global-truth access.

Navigation uses cheap deterministic operations such as:

- heading-to-point;

- cross-track error;
- bounded intercept angle;
- turn anticipation;
- altitude and speed error;
- formation offset;
- pattern generation; and
- precompiled route following.

All physical consequences remain owned by the applicable physical model.

AI does not set position, velocity, G, turn rate, weapon outcome, sensor detection, or control-surface state directly.

3. Player Modes - Realistic and Novice

Approved rewrite requirement / proposed gameplay contract

Mode	Intent	Effect
Realistic	Baseline behavior intended by this architecture	Normal doctrine thresholds, employment timing, defensive response, aggression, support discipline, and maneuver logic
Novice	More forgiving opposition	Same controller and tactics, but hostile employment occurs later, defensive reaction occurs later, commitment is reduced, and skate / abort behavior is favored earlier

3.1 Scope

Novice modifiers normally apply to **hostile tactical AI only**.

Friendly AI, including player wingman, AIC, and other Blue forces, remains governed by its normal doctrine unless Gameplay explicitly specifies otherwise.

3.2 Configuration-control treatment

Realistic / Novice use the same AI architecture:

same AI architecture
+ same state graphs
+ same maneuver primitives
+ modified hostile doctrine thresholds

Binding player-facing adoption, naming, defaults, menus, persistence, and any extension beyond hostile tactical AI require Gameplay / Main propagation.

4. Entity Classification and Doctrine Composition

4.1 Physical entity class versus mobility mode

An aircraft remains an **aircraft entity** for the entire mission instance.

Its physical implementation is fixed according to its approved aircraft physics class, such as the parent-defined SIX_DOF or approved KINEMATIC implementation.

Ground contact does not turn a SIX_DOF aircraft into a separate 2-D vehicle.

Instead, aircraft behavior carries a **mobility / contact mode** such as:

PARKED
GROUND_TAXI
TAKEOFF_ROLL
AIRBORNE
APPROACH
LANDING_CONTACT
ROLLOUT

A SIX_DOF aircraft remains within its physical aircraft/contact pipeline throughout these states.

Ships and ordinary ground vehicles remain surface-constrained kinematic entities.

Entity class	Mobility	Physical path
SIX_DOF aircraft	Ground and airborne	Aircraft physics + FCS + contact model
Approved KINEMATIC aircraft	Ground and airborne as applicable	Bounded kinematic aircraft controller subordinate to FlightDynamicsEngine
Ship / carrier	2-D surface	Surface kinematics
Ground vehicle / mobile SAM	2-D surface	Surface kinematics + precompiled route
Fixed SAM / fixed site	Static	No movement
Guided missile after launch	3-D weapon motion	WeaponAndDamageEngine, not AI movement

The KINEMATIC aircraft path is intentionally bounded. It remains subordinate to FlightDynamicsEngine and may not become a second uncontrolled physical-motion authority.

4.2 Combatant versus non-combatant

Combatant entities may detect, track, maneuver tactically, and engage under applicable ROE and capability.

Non-combatant entities follow authored routes, patterns, procedures, and safety logic only. They do not initiate tactical engagement.

4.3 Doctrine composition

Effective doctrine is composed rather than multiplied into a Cartesian package set:

```
effective_doctrine =  
    unit_base  
    + faction_modifiers  
    + proficiency_modifiers  
    + mission_posture_overrides  
    + optional_novice_hostile_modifiers
```

unit_base

Class-specific behavior:

- interceptor;
- strike aircraft;

- support aircraft;
- SAM battery;
- mobile SAM;
- surface combatant;
- carrier; or
- other approved class.

faction_modifiers

Influence:

- aggression;
- emission-control preference;
- support obligations;
- tactical conservatism;
- coordination doctrine.

Faction is not personality.

proficiency_modifiers

Rookie / Average / Veteran / Ace may affect:

- decision latency;
- maneuver precision;
- geometry estimation tolerance;
- timing;
- scan discipline;
- weapon discipline;
- coordination;
- support patience;
- commit and abort thresholds.

Proficiency never:

- changes physics;
- improves weapon performance;
- supplies hidden information; or
- changes sensor truth.

mission_posture_overrides

Examples:

- CAP;
- escort;
- intercept;
- strike;
- homeland defense;
- barrier;
- transit;
- recovery;
- support.

5. Bounded Perception and Knowledge

AI never cheats.

It consumes only declared information belonging to its tactical knowledge domain and its own physical/system state.

5.1 Legal AI knowledge

Permitted inputs include:

- semantic RadarTrack information from its legal knowledge domain;
- TrackQuality;
- own local fire-control state;
- legal weapon-support state;
- limited MEGA LINK / RED LINK information;
- RWR / MissileThreatState;
- own aircraft / vehicle physical and system state;
- OwnWeaponStatusView;
- WeaponEmploymentProjection;
- VisualContactView;
- mission-authored route information;
- RecoveryReference;
- facility clearance / occupancy information;
- formation coordination state.

Offboard tactical link may improve knowledge but **cannot** independently create local weapons quality or weapon-support validity.

5.2 OwnWeaponStatusView

WeaponAndDamageEngine supplies a bounded read-only AI projection.

Conceptually:

```
OwnWeaponStatusView {
    weapon_handle
    target_track_ref      // nullable / invalidatable
    support_required
    support_state
    autonomous
    active
    time_or_phase_class
    weapon_alive
}
```

target_track_ref is an AI association, not the weapon's authoritative target state.

If the associated semantic track is dropped or invalidated:

- the reference becomes invalid;
- it never silently binds to a newly allocated track using the same storage slot;
- weapon guidance remains Weapon-owned; and
- AI reassesses using currently legal knowledge.

SeekerAssessment is not ordinary AI knowledge.

5.3 WeaponEmploymentProjection

AI does not integrate missile trajectories to decide whether to shoot.

A bounded deterministic projection supplies tactical employment information.

Range geometry and effectiveness remain separate dimensions:

Dimension	Classes
Geometry	OUTSIDE, IN_ENVELOPE, NEAR, INSIDE_MIN
Estimated effectiveness	POOR, ACCEPTABLE, FAVORABLE, HIGH_EFFECT

Additional conceptual fields may include:

launch_acceptability
 own_weapon_timeline
 support_commitment_remaining
 target_escape_margin
 estimated_threat_weapon_zone
 abort_margin
 recommit_margin

The projection may use only:

- legal track estimates;
- legally known target classification;
- doctrine/intelligence assumptions;
- observed employment;
- own weapon data.

It may not expose:

- target truth state;
- unobserved enemy weapons;
- exact hostile fuel;
- exact hostile mass;
- hidden hostile configuration; or
- other private simulation state.

Any Pk-like value is an AI estimate, not an authoritative RNG kill probability.

5.4 VisualContactView

WVR behavior requires a declared visual-perception source.

VisualContactView, or the equivalent SensorAndTrack visual observation contract, supplies bounded legitimate visual information.

It may include observable:

- relative bearing;
- elevation;
- apparent range class;
- angular rate;
- aspect;
- visual classification;
- motion estimate.

Exact FOV, range, occlusion, cadence, and fidelity remain a later schema/tuning gate.

AI never obtains visual contact merely because the object exists in global truth.

5.5 IR-missile non-omniscience

The current F-65 has no dedicated IR missile-approach-warning system.

Therefore an AI aircraft without an authored IR-warning capability may not defend against an IR missile merely because WeaponAndDamageEngine contains that missile.

Legal IR warning sources may include only an approved modeled cue such as:

- visual missile observation;
- visual launch observation; or
- a platform-specific sensor actually installed on that entity.

Preemptive defensive flare doctrine is permitted where authored.

Secret access to incoming IR missile state is not.

This applies symmetrically to Blue and Red AI.

6. Hierarchical Doctrinal Controller

The complete controller hierarchy is:

HARD SAFETY / REFLEX

↓

MISSION

↓

FORMATION / TEAM

↓

OPERATIONAL / MOBILITY

↓

TACTICAL PHASE, when applicable

↓

MANEUVER / GUIDANCE

↓

CONTROL / INTENT OUTPUT

Tactical defense may preempt Mission, Formation, Operational, Tactical, and Maneuver behavior.

It may not preempt HARD SAFETY recovery.

Aviate-first or drive-first is absolute.

6.1 HARD SAFETY / REFLEX

Examples:

- terrain / ground collision;
- imminent aircraft collision;
- ship / ground-vehicle collision hazard;
- aircraft departure;
- loss of control;
- damage producing imminent flight-safety loss;
- surface hazard requiring immediate avoidance.

When HARD SAFETY owns the entity, only legal recovery behavior is permitted.

6.2 MISSION

Provides:

- authored objective;
- ROE;
- route;
- station;
- recovery destination;
- mission abort criteria.

AI does not invent strategic objectives.

6.3 FORMATION / TEAM

Coordinates:

- lead;
- formation;
- shooter/supporter roles;
- sorts;
- mutual support;
- rejoin;
- section abort;
- mission assignment.

Coordination never acts as a hidden sensor network.

6.4 OPERATIONAL / MOBILITY

Defines what the entity is physically doing:

- parked;
- starting;
- taxiing;
- launching;
- climbing;
- cruising;
- navigating;
- patrolling;
- holding;
- approaching;
- landing;
- rolling out;
- scooting;
- station keeping.

6.5 TACTICAL PHASE

Runs only when tactically applicable or when a defensive interrupt requires it.

Representative airborne offensive progression:

TRANSIT
→ SANITIZE
→ INTERCEPT
→ SORT
→ COMMIT
→ LAUNCH
→ SUPPORT
→ ASSESS
→ RECOMMIT / SECOND_SHOT
→ SKATE / ABORT
→ MERGE
→ BFM
→ SEPARATE
→ REJOIN

Defensive progression:

THREAT_AWARE
→ DEFEND
→ TERMINAL_DEFENSE
→ THREAT_LOST_STILL_DANGEROUS

- DEFEAT_CONFIRMED
- REASSESS

6.6 MANEUVER / GUIDANCE

Selects a bounded maneuver or navigation primitive.

6.7 CONTROL / INTENT OUTPUT

AI does not directly call forward into FCS, physics, weapons, sensors, or private module arrays. It produces the next-tick **AIIntentFrame** defined by the engine interface.

7. AIIntentFrame and Ownership Boundary

AIIntentFrame remains the parent stage-16 output contract.

Conceptually, the frame contains bounded intent classes such as:

```
AIIntentFrame {
    GuidanceIntent
    SystemIntent
    WeaponIntent
    SensorIntent
    coordination / mission intent as approved
}
```

Exact generated layout is an Engine/schema gate.

7.1 GuidanceIntent

Represents desired physical movement:

- course;
- altitude;
- speed/energy policy;
- formation offset;
- load-factor or lift-vector intent where legal;
- taxi/deck route;
- surface course;
- route segment;
- recovery guidance.

Consumers depend on entity type.

SIX_DOF aircraft

```
GuidanceIntent
  ↓
ControlAndSystemsEngine / FCS
  ↓
control mixer
  ↓
hydraulics / actuators
  ↓
actual surfaces
  ↓
FlightDynamicsEngine
```

KINEMATIC aircraft

GuidanceIntent

↓

bounded kinematic aircraft controller

↓

FlightDynamicsEngine

The kinematic controller remains subordinate to FlightDynamicsEngine and cannot mutate aircraft motion outside the approved kinematic physical contract.

Ship / ground vehicle

GuidanceIntent

↓

surface kinematic controller

↓

physical entity movement

A ship is never described as being controlled by an aircraft FCS.

7.2 SystemIntent

System commands remain separate from guidance.

Examples may include:

- engine start;
- gear;
- flaps;
- hook;
- brakes;
- ADLC request;
- approved system configuration.

They are consumed by their owning systems/control module.

A guidance record does not directly manipulate aircraft-system state.

7.3 WeaponIntent

Weapon selection, release, support, or other tactical weapon requests remain owned by the weapon interface.

WeaponAndDamageEngine retains physical launch legality and the weapon lifecycle.

7.4 SensorIntent

Radar mode, scan, emission, or other declared sensor requests remain owned by SensorAnd-Track/system interfaces.

AI does not manipulate sensor internals.

8. Held Guidance Semantics and Decision Cadence

AI does not need to regenerate low-level navigation commands at the 100 Hz physics rate.

A stage-16 guidance output is a **held intent**.

Conceptually:

stage 16, tick N:
 AI updates GuidanceIntent

tick N+1 onward:
 legal consumer latches new intent

subsequent 100-Hz ticks:
 controller continues closing against held intent

later stage-16 decision:
 replace, cancel, or modify intent

Examples:

```
CLIMB_TO 28,000 ft
TURN_TO course 090
HOLD 350 KIAS
FOLLOW formation offset
TRACK 3.5-degree recovery glidepath
SHIP course 270 / speed policy TRANSIT
FOLLOW ground RouteSegment 12
```

The high-rate owning controller performs the continuous physical correction.

This is distinct from AI deliberation.

The exact generated lifetime, validity-mask, cancel, replacement, and invalidation semantics remain an Engine/schema gate. The durable behavioral requirement is that a legal held intent persists until completion, replacement, cancellation, invalidation, mode transition, or HARD SAFETY override.

9. Mobility, Navigation, and Procedural Operations

This is the seventh AI pillar.

Submodel	Principle
Air navigation	Authored RoutePlan + cheap guidance
Aircraft procedural operation	Bounded parking-to-parking FSM
Aircraft taxi/deck movement	Precompiled taxi/deck routes
Ship navigation	Authored sea routes and patterns
Ground vehicle navigation	Precompiled routes and relocation choices

No general runtime pathfinding is permitted.

10. Air Navigation and Energy Management

10.1 RoutePlan

A mission-authored bounded route contains prevalidated legs.

Conceptually:

```
RoutePlan {
  RouteLeg legs[bounded_count]
  current_leg
  suspended_leg
```

```
    alternate_route_id  
}
```

A RouteLeg may contain:

```
reference  
leg_type  
altitude_target_or_band  
speed_policy  
arrival_radius  
next_action
```

Representative leg types:

```
DIRECT  
HEADING  
CLIMB  
DESCEND  
ORBIT  
RACETRACK  
HOLD  
FORMATION  
INTERCEPT  
APPROACH  
RECOVERY
```

Runtime steering may use:

- heading-to-waypoint;
- cross-track error;
- bounded course intercept;
- turn anticipation;
- altitude error;
- speed error;
- arrival-radius test.

10.2 Deterministic route interruption and resumption

The phrase “recompute route” does not authorize runtime path planning.

When tactical or defensive behavior interrupts navigation:

preserve suspended_route_leg

When the override ends:

1. If the suspended/current route remains legal, rejoin it or its authored successor.
2. If it is invalid because of mission state, use an **authored alternate / divert / recovery route**.
3. If no legal authored continuation exists, enter the applicable hold / abort / safe-state behavior.

AI may not generate a new route graph.

Example:

```
WP4  
→ INTERCEPT / COMBAT  
→ SEPARATE  
→ REJOIN_ROUTE  
→ WP5
```

or:

```
ROUTE  
→ DAMAGE  
→ authored DIVERT_ROUTE_2
```

10.3 Guidance primitives

Representative primitives:

TURN_TO_COURSE
CLIMB_TO_ALTITUDE
DESCEND_TO_ALTITUDE
HOLD_ALTITUDE
ACCELERATE_TO_ENERGY
DECELERATE_TO_SPEED
MAINTAIN_SPEED
FOLLOW_ROUTE
ORBIT_POINT
FOLLOW_FORMATION

These are intent-generation primitives, not alternate physics.

10.4 Speed and energy policy

AI never sets physical velocity directly.

Examples:

ECONOMY_CRUISE
TACTICAL_INGRESS
COMBAT_CRUISE
MAX_ACCEL
RECOVERY_DECEL
ON_SPEED_APPROACH

The underlying physical model determines:

- acceleration;
- spool behavior;
- fuel consumption;
- drag;
- attainable speed;
- climb performance.

afterburner_authorization permits AB where doctrinally appropriate; it does not create immediate thrust.

10.5 Formation navigation

Lead follows the route or tactical objective.

Wingmen maintain authored relative geometry using bounded formation references.

Formation knowledge does not convey hidden tactical tracks.

11. Aircraft Ground and Flight Procedures

Aircraft procedural operation covers the full mission.

Representative operational FSM:

PARKED / COLD
→ START / READY
→ TAXI
→ HOLD / LINEUP
→ TAKEOFF / CATAPULT

- INITIAL_CLIMB
- CLEANUP
- DEPARTURE
- CLIMB
- CRUISE / NAVIGATE
- TACTICAL_INGRESS
- COMBAT
- EGRESS
- RETURN_NAV
- DESCENT
- RECOVERY_ENTRY
- APPROACH
- FINAL
- LAND / ARREST
- ROLLOUT
- TAXI
- PARK

11.1 Cold versus already-running mission start

An AI aircraft may begin a mission either:

1. explicitly authored **already operating / ready**, or
2. authored **cold**, in which case readiness must be reached through the normal system model.

A cold-start procedural abstraction may be:

COLD

- POWER_AVAILABLE
- ENGINE_START
- ENGINE_STABLE
- GENERATION / HYDRAULIC_CAPABILITY
- AVIONICS / REQUIRED_SYSTEMS_READY
- TAXI_READY

Exact switch-by-switch fidelity is not required for AI unless Gameplay later requires it.

AI may not set physical engine/system readiness instantaneously merely because the next procedural state is TAXI.

11.2 Airfield TaxiRoute

Airfield taxi uses an authored/precompiled route.

Conceptually:

```
TaxiRoute {
    start_reference
    TaxiSegment segments[bounded_count]
    hold_short_reference
    runway_entry_reference
}
```

Runtime behavior:

PARK

- FOLLOW_TAXI_SEGMENT
- TURN
- HOLD_SHORT
- runway clearance
- LINE_UP

No taxiway path search occurs during the sortie.

11.3 Carrier DeckRoute

Carrier movement uses equivalent authored guidance:

```
DeckRoute {  
    parking_reference  
    deck_segments[bounded_count]  
    queue_reference  
    catapult_assignment  
    catapult_alignment_reference  
}
```

Runtime behavior:

```
PARK  
→ DECK_TAXI  
→ CAT_QUEUE  
→ CAT_ALIGN  
→ HOLD / HOOKUP
```

The route moves with the carrier-local frame.

11.4 Airfield takeoff

```
HOLD_SHORT  
→ LINE_UP  
→ TAKEOFF_ROLL  
→ ROTATE  
→ POSITIVE_CLIMB  
→ GEAR_UP  
→ FLAP_RETRACT  
→ INITIAL_CLIMB  
→ DEPARTURE
```

Rotation is based on the aircraft's legal weight/configuration performance schedule.

AI waits for actual aircraft state.

It does not:

- force liftoff;
- set a vertical velocity;
- teleport airborne; or
- bypass ground/contact physics.

11.5 Carrier launch

```
DECK_TAXI  
→ CAT_QUEUE  
→ CAT_ALIGN  
→ HOOKUP / READY  
→ LAUNCH_CONFIGURATION  
→ LAUNCH  
→ FLYAWAY  
→ CLEANUP
```

Catapult launch remains a physical launch/contact event owned by the applicable simulation systems.

AI requests readiness and launch participation; it does not inject catapult acceleration itself.

11.6 Configuration sequencing

AI obeys the same physical configuration limits as the player.

Example:

```
RECOVERY_ENTRY
→ DECELERATE
→ gear legal?
→ GEAR_DOWN
→ flap setting legal?
→ FLAPS
→ HOOK_DOWN if carrier
→ APPROACH_CAPTURE
```

Configuration requests occur through `SystemIntent`.

Gear, flap, hook, hydraulic, actuator, and damage behavior remain system-owned.

11.7 RecoveryReference

AI recovery guidance uses geometric world information rather than rendered pixels.

Conceptually:

```
RecoveryReference {
    facility_reference
    threshold_or_touchdown_reference
    landing_course
    carrier_local_transform
    glidepath_reference
    lineup_reference
}
```

Derived errors may include:

```
lineup_error
glidepath_error
AoA_or_speed_error
sink_rate
distance_to_touchdown
```

These are navigation/guidance quantities.

They do not guarantee a successful landing.

Physical contact remains authoritative.

11.8 Airfield visual recovery

Representative states:

```
RECOVERY_ENTRY
→ DESCENT
→ PATTERN_ENTRY / STRAIGHT_IN
→ CONFIGURE
→ FINAL_APPROACH
→ STABILIZED_APPROACH_CHECK
```

If stabilized:

```
CONTINUE
→ LANDING_TRANSITION / FLARE
→ TOUCHDOWN
→ ROLLOUT
```

→ RUNWAY_EXIT
→ TAXI
→ PARK

If unstable:

GO_AROUND
→ positive climb
→ configuration cleanup as physically legal
→ CLIMB
→ REENTER_PATTERN

or, if required:

DIVERT

Stabilized-approach criteria are authored/tunable and may include:

- lineup error;
- glidepath error;
- sink rate;
- speed/AoA;
- configuration;
- system capability;
- facility occupancy;
- available runway geometry.

AI is not required to continue an unsafe approach merely because touchdown remains geometrically possible.

11.9 Carrier recovery

Representative simplified Case I sequence:

RECOVERY_ENTRY
→ CASE_I_ENTRY
→ INITIAL
→ BREAK
→ DOWNWIND
→ ABEAM
→ 90
→ FINAL
→ GROOVE

Within the final/groove:

continue if recoverable
or
WAVEOFF if approach becomes unacceptable

At deck contact:

wire captured
→ ARRESTED

or:

no wire
→ BOLTER
→ CLIMB
→ REENTER_PATTERN

A bolter is a physical contact outcome.

A waveoff is an AI procedural decision before an unsafe touchdown attempt.

11.10 ADLC

When available and appropriately configured, AI may request ADLC operation through the declared systems/control path.

ADLC retains its parent-defined physical behavior.

If unavailable or degraded, doctrine selects among:

CONVENTIONAL_APPROACH
GO_AROUND / WAVEOFF / BOLTER
DIVERT

according to capability and mission state.

12. Launch, Recovery, and Facility Coordination

Multiple AI aircraft require bounded deterministic arbitration for shared facilities.

This is not a general ATC simulation.

A facility-owned projection or mission/service record may expose concepts such as:

```
FacilityClearanceView {  
    runway_available  
    runway_occupied  
    launch_clearance  
    landing_clearance  
    catapult_assignment  
    catapult_available  
    recovery_sequence  
    deck_or_runway_hold  
}
```

Exact schema and ownership are Engine / Mission / facility-interface work.

The required behavioral rule is:

- AI does not enter an occupied or unauthorized runway/catapult/recovery slot merely because its procedural FSM reached the next state.
- HOLD states remain legal indefinitely until clearance or an interrupt changes the mission.
- Ordering and ties are deterministic.
- Facility state does not grant hidden tactical knowledge.

Examples:

```
TAXI  
→ HOLD_SHORT  
→ wait for runway_available + launch_clearance  
→ LINE_UP  
  
CAT_QUEUE  
→ wait for catapult_assignment / clearance  
→ CAT_ALIGN  
  
RECOVERY_ENTRY  
→ recovery_sequence  
→ pattern / final
```

13. Ship Navigation

Ships remain surface-kinematic entities.

Representative states:

MOORED
HOLD
TRANSIT
PATROL
RACETRACK
STATION_KEEP
FORMATION
TACTICAL_MANEUVER
RETURN_ROUTE

An authored ship route contains bounded legal water legs.

Runtime navigation uses:

desired_course
desired_speed
next_waypoint
cross_track_error
turn_rate_limit
station_offset

A carrier may use an authored recovery course or pattern when recovery operations require it.

Warships may temporarily depart their route:

TRANSIT
→ TACTICAL_MANEUVER
→ REJOIN_ROUTE

No runtime maritime pathfinding occurs.

Local reflex logic may avoid:

- collision with another live vessel;
- obviously invalid immediate course;
- other approved immediate surface hazards.

Mission authoring is responsible for normal route validity.

14. Ground-Vehicle Navigation

Ground vehicles do not solve arbitrary terrain paths at runtime.

Mission compilation creates a bounded route network:

RouteNode
RouteSegment
FiringPosition
HoldPoint

Runtime states may include:

PARKED
→ MOVE_ROUTE
→ FOLLOW_SEGMENT
→ TURN_AT_NODE

- ARRIVE
- HOLD / EMPLACE

14.1 Precompiled relocation options

A mobile SAM commander may not merely select a destination and then search the route graph.

Each legal relocation choice contains its already-compiled path.

Conceptually:

```
RelocationOption {  
    destination_site  
    route_id  
}
```

A firing site contains a bounded list:

```
Site_B:  
    option → Site_C via Route_05  
    option → Site_D via Route_07  
    option → Site_F via Route_09
```

Selection may use doctrine and, where authorized, the entity-local deterministic AI RNG.

Once selected:

```
DARK  
→ PACK  
→ SCOOT via selected route_id  
→ ARRIVE  
→ EMPLACE
```

No graph search occurs.

14.2 Local ground avoidance

A small reflex may prevent immediate collision or illegal movement.

It may:

- stop;
- slow;
- wait;
- take an already-authored local alternate segment.

It may not invent an arbitrary off-network route.

15. Mission Navigation Override and Resume Contract

Across all mobility domains:

```
authored route / pattern  
→ tactical or defensive override  
→ deterministic authored continuation
```

Examples:

Aircraft

ROUTE LEG 5
→ INTERCEPT
→ COMBAT
→ SEPARATE
→ REJOIN ROUTE LEG 6

Ship

PATROL LEG 3
→ TACTICAL_MANEUVER
→ RETURN_ROUTE

Mobile SAM

EMPLACED SITE B
→ ENGAGE
→ DARK
→ RelocationOption B→D / Route_07
→ SITE D

The AI never treats “return to mission” as permission for general route recomputation.

16. Deterministic Interrupt Hierarchy

Interrupt priority is based on consequence.

16.1 Level 1 - FLIGHT / SURFACE SAFETY

Examples:

- terrain collision;
- imminent collision;
- departure;
- loss of control;
- hydraulic/system damage causing immediate control loss;
- dangerous surface motion.

Always wins.

16.2 Level 2 - TERMINAL SURVIVAL

A credible, legally cued missile threat requiring immediate defense.

It preempts normal mission/tactical work but not flight-safety recovery.

16.3 Level 3 - TACTICAL

Examples:

- fire-control attention;
- deteriorating combat geometry;
- wingman emergency;
- tactically significant sensor event.

16.4 Level 4 - MISSION LIMIT

Examples:

- Bingo / fuel;
- survivable damage;
- mission abort;
- recovery requirement.

A damage event may migrate to Level 1 if its consequence becomes immediate loss of aircraft control.

17. Reflex Pass and Doctrine Pass

All AI execution remains at stage 16.

There is no secondary authoritative simulation clock.

REFLEX PASS

Runs on every eligible AI evaluation for the entity.

Handles:

- immediate safety;
- collision avoidance;
- departure recovery;
- legally cued missile threat;
- immediate defensive transitions;
- local surface avoidance.

DOCTRINE PASS

May run at a lower scheduled cadence.

Handles:

- target selection;
- sorting;
- commit/recommit;
- formation;
- mission-limit evaluation;
- BVR local utility;
- IADS emission planning;
- route sequencing;
- operational-state changes;
- recovery planning.

Continuous physical control between decisions is provided by the owning control/guidance system using the held intent from Section 8.

18. BVR Weapon Employment

WeaponAndDamageEngine owns:

- physical weapon state;
- launch legality;
- guidance;

- seeker state;
- support consumption;
- missile physics;
- fuze;
- damage.

AI owns:

- tactical weapon choice;
- tactical launch decision;
- shot doctrine;
- support exposure;
- crank/press/drag decision;
- follow-on shot timing;
- abort/recommit decision.

Representative BVR phases:

INTERCEPT

- SORT
- COMMIT
- LAUNCH
- SUPPORT
- ASSESS
- SECOND_SHOT / RECOMMIT / SKATE / ABORT / MERGE

After launch, doctrinally legal options might be:

PRESS
CRANK_LEFT
CRANK_RIGHT
DRAG
ABORT

A bounded fixed-point utility evaluation ranks only these legal alternatives.

Possible inputs:

- support time remaining;
- own missile status;
- inbound threat;
- target value;
- fuel margin;
- wingman state;
- predicted crank geometry;
- abort margin.

Stable ties use a frozen deterministic tie-break.

19. Defensive Controller

Representative defensive graph:

THREAT_AWARE
→ DEFEND
→ TERMINAL_DEFENSE
→ THREAT_LOST_STILL_DANGEROUS
→ DEFEAT_CONFIRMED
→ REASSESS

Legal maneuver/countermeasure actions may include:

- beam / notch;
- drag;
- break;
- unload / accelerate;
- terrain masking;
- chaff;
- flare;
- jammer;
- doctrinal combinations.

No defensive action receives a magical success bonus.

Notching remains dependent on the actual SensorAndTrack model.

19.1 F-65 jammer trade-off

The defensive jammer is not free.

When applicable under the Sensor baseline it may:

- blank own radar;
- deny MEGA LINK transmit;
- deny MEGA LINK receive;
- invalidate radar-dependent missile support at the legal consumption point.

Therefore AI may face a legitimate choice:

continue supporting own missile

versus

abandon support and jam defensively

This is resolved through doctrine and survival priority, not special-case cheating.

20. WVR / BFM

BFM uses bounded geometry classification and maneuver primitives.

Representative primitives include:

MERGE_ASSESS
 RATE
 RADIUS
 HIGH_YOYO
 LOW_YOYO
 LAG_PURSUIT
 PURE_PURSUIT
 LEAD_PURSUIT
 EXTEND
 REVERSE
 DEFENSIVE_BREAK
 UNLOAD_ACCELERATE
 SEPARATE

20.1 Relational fight topology

An AI does not unilaterally select “one-circle” or “two-circle.”

Instead:

MERGE_ASSESS

- desired turn / fight intent
- both aircraft maneuver physically
- observe resulting topology
- rate / radius / pursuit strategy

20.2 Energy estimation

AI knows its own aircraft state exactly.

Opponent energy is estimated from legal information only.

own_energy_state
= own physical aircraft state

opponent_energy_estimate
= legal track / VisualContactView kinematics

AI does not inspect enemy:

- fuel;
- exact weight;
- exact thrust state;
- hidden stores; or
- exact specific excess power.

20.3 Performance-class comparison

Own aircraft uses its exact authored maneuver-performance class.

Opponent performance class is estimated only from legal identification.

A tiny bounded table informs whether doctrine should prefer:

- sustained rate;
- minimum radius;
- vertical maneuver;
- extension;
- rapid angular conversion.

The AI does not simulate a second copy of the opponent's full 6DOF performance model.

The actual aircraft still pays the full physical energy cost.

21. IADS Doctrine

A battery/network may use roles such as:

EARLY_WARNING
ACQUISITION
TRACK
ENGAGEMENT
PASSIVE / AMBUSH
RELOCATION

Emission policy depends on:

- radar role;
- sensor profile;
- network knowledge;
- mission doctrine;

- current threat;
- engagement requirement.

No simplistic rule states that “2-D radar is quiet” or “3-D radar produces an RWR spike.”

Representative sequence:

PASSIVE

- legal RED LINK / local cue
- SEARCH_CUE
- acquisition emission
- TRACK_CONFIRMED
- engagement sensor
- ENGAGE
- MISSILE_SUPPORT
- DARK
- SCOOT if mobile

The network commander coordinates only legally available network information.

It is not an omniscient RED truth controller.

22. Ship Tactical Doctrine

Equipped surface combatants may execute:

SEARCH / LINK SA

- CLASSIFY
- THREAT RANK
- ENGAGE AIRCRAFT if equipped
- ENGAGE INBOUND WEAPON if equipped
- SELF DEFENSE
- EMCON / JAM / MANEUVER

Scarce channels, launchers, sensors, and weapons are allocated deterministically.

A ship does not fire every compatible weapon at the first detected target merely because a weapon is available.

Capabilities exist only where current sensor/weapon profiles define them.

23. Skill, Doctrine, and Variation

Skill changes decision quality, not physical reality.

Proficiency may alter:

- reaction latency;
- transition thresholds;
- maneuver timing;
- maneuver precision;
- scan/emission discipline;
- tactical prediction tolerance;
- shot discipline;
- support patience;
- abort willingness;
- formation coordination.

It may not alter:

- flight physics;
- weapon kinematics;
- seeker performance;
- radar performance;
- damage;
- hidden information access.

Any authorized variability uses the entity-local AI RNG stream only.

24. Deterministic Implementation Model

The intended implementation is small, table-driven, and bounded.

AI runtime state may contain:

- current hierarchy layer;
- operational state;
- tactical phase;
- interrupt state;
- route leg/node;
- held guidance intent;
- target reference;
- team role;
- weapon-status summary;
- limited candidate set.

Doctrine resides in compact authored/generated tables.

Required implementation constraints:

- fixed capacities;
 - no heap;
 - no recursion;
 - no target floating point;
 - no runtime path search;
 - bounded candidate lists;
 - fixed iteration limits;
 - stable tie-breaks;
 - entity-local AI RNG only where authorized;
 - C first;
 - handwritten 45GS02 assembly only when admitted by parent architecture or measurement.
-

25. Memory, Code, Data, and Cycle Measurement

The AI/mobility architecture must be measured as a complete owner, not only by average CPU time.

R0 / Phase-1 evidence shall identify at minimum:

25.1 Resident code

- AI core code bytes;
- mobility/navigation code bytes;
- formation/team code bytes;
- mission/doctrine runtime bytes;
- any compiler/runtime support charged to this owner.

These must fit the applicable Engine code ledger, including the existing approximately 5 KB AI/formations/mission owner budget unless the controlled measured-limits process changes that budget.

25.2 Runtime state

Measure:

- per-aircraft AI state;
- per-surface-entity AI state;
- per-flight tactical state;
- IADS/battery state;
- held intent storage;
- facility coordination state;
- route cursors;
- diagnostic state.

25.3 Authored / mission-hot data

Measure:

- RoutePlan storage;
- RouteLeg storage;
- airfield TaxiRoute;
- carrier DeckRoute;
- ground RouteNode / RouteSegment data;
- RelocationOption tables;
- RecoveryReference;
- doctrine tables;
- performance comparison tables.

Host-generated/cold mission data and live resident bytes shall be reported separately.

25.4 CPU evidence

Measure legal worst cases for:

- reflex pass;
- doctrine pass;
- route sequencing;
- formation update;
- BVR target/utility evaluation;
- BFM selection;
- IADS planning;
- local avoidance;
- procedural phase transition.

The combined-load case must exercise the legal overlap of reflex + doctrine + mobility work. Average cost cannot replace worst legal combined-load evidence.

26. Debugging and Diagnostics

A bounded diagnostic view should expose sufficient state to understand why an AI acted.

Suggested logical diagnostics:

entity
current hierarchy owner

interrupt class
mission goal
operational / mobility phase
current route / leg / node
suspended route
tactical phase
primary TacticalTrackRef
formation/team role
OwnWeaponStatusView summary
VisualContactView presence
current held GuidanceIntent
SystemIntent summary
performance-class comparison
last local utility winner
facility clearance / hold state

Diagnostics remain read-only and cannot change simulation.

27. Post-Freeze Implementation and Propagation Gates

The following are **natural next gates after conceptual freeze**. They do **not** reopen or withhold approval of Version 1.0.

27.1 Engine and generated-schema closure

Engine/schema work must close the exact generated layouts and interface details for:

- AIIntentFrame;
- GuidanceIntent modes and validity masks;
- SystemIntent;
- WeaponIntent;
- SensorIntent;
- held-intent lifetime, cancel, replacement, and invalidation semantics;
- RoutePlan / RouteLeg;
- TaxiRoute;
- DeckRoute;
- RecoveryReference;
- FacilityClearanceView;
- VisualContactView;
- OwnWeaponStatusView; and
- WeaponEmploymentProjection.

These are generated-layout and public-interface tasks, not unresolved conceptual AI architecture.

27.2 Degraded-FCS mapping under Physics v3.3

Physics / FCS integration must provide the deterministic table or mapping from AI guidance modes to the legal degraded control behavior under:

- FCSStatus = NORMAL;
- FCSStatus = DEGRADED; and
- FCSStatus = DIRECT_ONLY.

The conceptual requirement is already frozen here: AI may request only behavior the remaining control capability can honor and may never bypass the physical FCS/control path.

27.3 Bounded KINEMATIC aircraft path

The parent Physics/Engine model already contemplates an approved KINEMATIC aircraft class.

Its implementation must remain:

- fixed at mission load;
- bounded;
- deterministic;
- subordinate to FlightDynamicsEngine;
- compatible with the same lifecycle, collision, sensing, and 100 Hz physical timeline; and
- incapable of becoming a second uncontrolled motion path outside the engine ownership model.

27.4 Gameplay / Main propagation

The coordinated 1.6 rewrite must resolve and record:

- adoption of Realistic / Novice;
- hostile-only Novice scope unless Gameplay deliberately expands it;
- default mode;
- menu/settings exposure;
- persistence behavior; and
- formal listing of this AI paper as the fifth detailed subsystem white paper if that is the intended corpus shape.

27.5 R0 / Phase-1 measurement of the full AI owner

The complete AI owner must be measured against the Engine ledger, including:

- resident code bytes;
- per-entity state bytes;
- per-flight / battery / facility state;
- route and doctrine data;
- generated tables;
- stack/scratch impact where applicable;
- worst-case reflex cost;
- worst-case doctrine cost;
- worst-case mobility/procedural cost; and
- legal combined-load reflex + doctrine + mobility execution.

The measured result, not source-code estimates or average timing, determines final budget closure.

28. Summary Contract - Version 1.0

1. F-65 AI uses a bounded **hierarchical doctrinal controller**, not neural inference or general planning.

2. The hierarchy is:

HARD SAFETY

→ MISSION

→ FORMATION / TEAM

→ OPERATIONAL / MOBILITY

→ TACTICAL

→ MANEUVER / GUIDANCE

→ CONTROL / INTENT OUTPUT

3. HARD SAFETY always outranks tactical behavior.

4. Reflex evaluation handles immediate safety and legally cued threats; lower-cadence doctrine evaluation handles tactical and operational decisions.
5. AI knowledge is strictly bounded. No global truth, no sensor telepathy, no IR-missile omniscience, and no hidden enemy inventory.
6. AIIntentFrame remains the stage-16 parent output. AI does not directly control private FCS, systems, weapons, sensors, or surface-motion state.
7. Guidance, systems, weapons, and sensors remain distinct ownership domains.
8. Guidance targets are **held intents**. High-rate owning controllers continue closing against the latest legal target between AI decisions.
9. Aircraft remain aircraft throughout ground and airborne operations. Ground taxi, takeoff roll, airborne flight, landing contact, and rollout are mobility/contact states, not a permanent physical-domain switch.
10. Air navigation uses authored RoutePlan legs and cheap fixed-point guidance. No general runtime pathfinding exists.
11. Taxi and carrier-deck navigation use authored/precompiled routes.
12. Aircraft startup is either an explicit already-running mission condition or a bounded procedure through the real system model.
13. Takeoff, catapult launch, acceleration, climb, configuration, landing, arrestment, bolter, and rollout remain physical.
14. Airfield recovery includes stabilized-approach evaluation and go-around. Carrier recovery includes waveoff and physical bolter behavior.
15. Shared runway/catapult/recovery access uses bounded deterministic facility-clearance state rather than uncontrolled simultaneous use.
16. Ships follow authored water routes/patterns, may perform tactical deviations, and return through authored route state.
17. Ground vehicles follow precompiled paths. Mobile-SAM relocation choices include both destination and precompiled route; selecting a site never triggers runtime graph search.
18. Mission-route interruptions preserve a suspended route and resume through authored continuation, alternate, divert, or recovery routes only.
19. BVR AI reasons about geometry, estimated weapon effectiveness, support timelines, crank/press/drag behavior, abort and recommit without integrating weapon physics itself.
20. Defensive behavior uses real sensor/threat geometry and countermeasure rules. The jammer's cost to own radar/link/support is a real tactical trade-off.
21. BFM topology is relational. Rate/radius preference uses legal geometry, estimated opponent energy, and bounded aircraft performance-class comparison.
22. IADS uses role-based emission control, legal shared knowledge, ambush behavior, engagement support, darkness, and precompiled shoot-and-scoot mobility.
23. Skill changes reaction, precision, doctrine, and decision quality. It never changes physics or grants hidden knowledge.
24. Realistic / Novice use the same controller. Novice is an approved rewrite requirement for hostile-AI difficulty behavior pending the Gameplay / Main propagation details in Section 27.4.
25. Authoritative AI is fixed-capacity, fixed-point, deterministic, non-recursive, and pathfinding-free.
26. The approved KINEMATIC aircraft path remains bounded and subordinate to FlightDynamicsEngine; it cannot become an uncontrolled second motion authority.
27. AI decisions remain at stage 16 or slower and produce effects no earlier than the following tick.

28. Code bytes, runtime state bytes, mission-hot route/doctrine data, and worst-case cycles are all measured against the Engine and measured-limits ledgers.
29. Generated interface layouts, degraded-FCS tables, KINEMATIC implementation details, Gameplay/Main propagation, and R0/Phase-1 measurements are post-freeze implementation gates. They do not reopen this conceptual approval unless evidence exposes an actual contradiction with a controlling parent.
30. **Version 1.0 is the human-reviewed AI Behavior and Decision Architecture engineering baseline and approved rewrite input for the subsequent F-65 1.6 specification set.**

End of F-65 Megawing AI Behavior and Decision Architecture White Paper v1.0